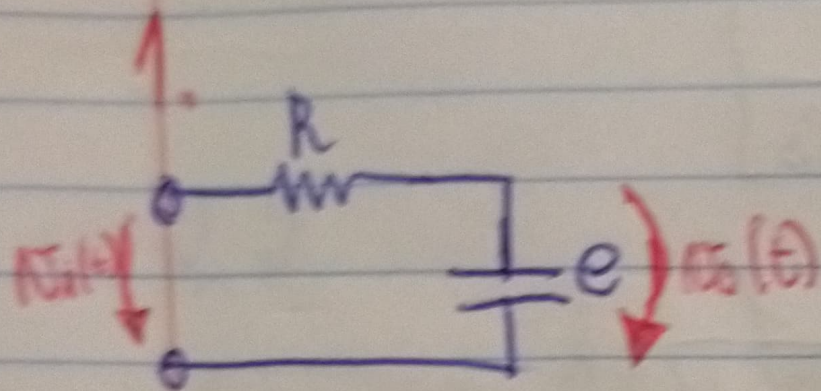


Exercício 3 (Resposta Frequencial 1.º ordem)



$$R = 1 \text{ k}\Omega$$

$$C = 1 \mu\text{F}$$

$$v_i(t) = A \sin(\omega t)$$

$$RC \frac{dv_o(t)}{dt} + v_o(t) = v_i(t)$$

$$N_0(t) = N_i(t) - \operatorname{Re} \frac{dN_0(t)}{dt}$$

$$\Rightarrow \mathcal{L}\{N_0(t)\} = \mathcal{L}\{N_i(t)\} - \mathcal{L}\left\{\operatorname{Re} \frac{dN_0(t)}{dt}\right\}$$

$$\Rightarrow V_0(s) = \mathcal{L}\{A \sin(\omega t)\} - \operatorname{Re} \mathcal{L}\left\{\frac{dN_0(t)}{dt}\right\}$$

$$\Rightarrow V_0(s) = A \mathcal{L}\{\sin(\omega t)\} - \operatorname{Re}(s V_0(s) - N_0(0))$$

$$\Rightarrow V_0(s) + \operatorname{Re} s V_0(s) - \operatorname{Re} N_0(0) = \frac{A\omega}{s^2 + \omega^2}$$

$$\Rightarrow V_0(s)(1 + \operatorname{Re} s) = \frac{A\omega}{s^2 + \omega^2} + \operatorname{Re} N_0(0)$$

$$\Rightarrow V_0(s) = \frac{A\omega}{(1 + \operatorname{Re} s)(s^2 + \omega^2)} + \frac{\operatorname{Re} N_0(0)}{1 + \operatorname{Re} s}$$

$$\Rightarrow N_0(t) = \mathcal{L}^{-1}\left\{\frac{A\omega}{(1 + \operatorname{Re} s)(s^2 + \omega^2)}\right\} + \mathcal{L}^{-1}\left\{\frac{\operatorname{Re} N_0(0)}{1 + \operatorname{Re} s}\right\}$$

$$\Rightarrow N_0(t) = A\omega \mathcal{L}^{-1}\left\{\frac{1}{(1 + \operatorname{Re} s)(s^2 + \omega^2)}\right\} + \operatorname{Re} N_0(0) \mathcal{L}^{-1}\left\{\frac{1}{1 + \operatorname{Re} s}\right\}$$

$$\Rightarrow N_0(t) = A\omega \mathcal{L}^{-1}\left\{\frac{1}{\operatorname{Re}(s + \frac{1}{\operatorname{Re}})(s^2 + \omega^2)}\right\} + \operatorname{Re} N_0(0) \mathcal{L}^{-1}\left\{\frac{1}{\operatorname{Re}(s + \frac{1}{\operatorname{Re}})}\right\}$$

$$\Rightarrow N_0(t) = \frac{A\omega}{\operatorname{Re}} \mathcal{L}^{-1}\left\{\frac{1}{(s + \frac{1}{\operatorname{Re}})(s^2 + \omega^2)}\right\} + \frac{\operatorname{Re} N_0(0)}{\operatorname{Re}} \mathcal{L}^{-1}\left\{\frac{1}{s + \frac{1}{\operatorname{Re}}}\right\}$$

$$\Rightarrow N_0(t) = \frac{A\omega}{\operatorname{Re}} \left(\frac{e^{-\frac{1}{\operatorname{Re}}t}}{(\frac{1}{\operatorname{Re}})^2 + \omega^2} + \frac{1}{\omega \sqrt{(\frac{1}{\operatorname{Re}})^2 + \omega^2}} \sin(\omega t - \phi) \right) + N_0(0) e^{-\frac{1}{\operatorname{Re}}t}$$

$$\Rightarrow N_0(t) = \frac{A\omega}{\operatorname{Re}} \frac{e^{-\frac{1}{\operatorname{Re}}t}}{(\frac{1}{\operatorname{Re}})^2 + \omega^2} + \frac{A \sin(\omega t - \phi)}{\operatorname{Re} \sqrt{(\frac{1}{\operatorname{Re}})^2 + \omega^2}} + N_0(0) e^{-\frac{1}{\operatorname{Re}}t}$$

2.

2.1.

$$g(\omega) = \frac{K_{af}}{\sqrt{\left(\frac{\omega}{\omega_0}\right)^2 + 1}}$$

$$\omega_0 = \frac{1}{\tau} = \frac{1}{R_e}$$

$$\lim_{t \rightarrow \infty} N_0(t)$$

$$\tau = R_e$$

$$= \lim_{t \rightarrow \infty} \frac{A\omega}{R_e \sqrt{\left(\frac{\omega}{\omega_0}\right)^2 + 1}} e^{\frac{t}{R_e}} + \frac{A \sin(\omega t - \phi)}{R_e \sqrt{\left(\frac{\omega}{\omega_0}\right)^2 + 1}} + N_0(0) e^{-\frac{t}{R_e}}$$

$$\Rightarrow \tau = 1000 \times 1 \times 10^{-6}$$

$$\Rightarrow \tau = 1 \times 10^{-3} \text{ s}$$

$$= \lim_{t \rightarrow \infty} \frac{A \sin(\omega t - \phi)}{R_e \sqrt{\omega^2 + \left(\frac{\omega}{\omega_0}\right)^2}}$$

$$= \frac{A}{R_e \sqrt{\omega^2 + \left(\frac{\omega}{\omega_0}\right)^2}}$$

2.2.

$$\phi = -\arctan\left(\frac{\omega}{\omega_0}\right)$$

$$\tan(\phi) = \frac{\omega}{\omega_0}$$

2.3.

A onda vai ser desfasada e a amplitude menor da onda de saída.

2.4.

$$\phi = -45^\circ$$

$$\phi = -\arctan\left(\frac{\omega}{\omega_0}\right)$$

$$\Rightarrow \phi = -\arctan(\omega R E)$$

$$\Rightarrow -45 = -\arctan(\omega \times 1 \times 10^{-3})$$

$$\Rightarrow 45 = \arctan(\omega \times 1 \times 10^{-3})$$

$$\Rightarrow 1 = 1 \times 10^{-3} \omega$$

$$\Rightarrow \omega = 1000 \text{ rad/s} \quad \rightarrow \text{Corresponde à constante de tempo do circuito } (\tau)$$

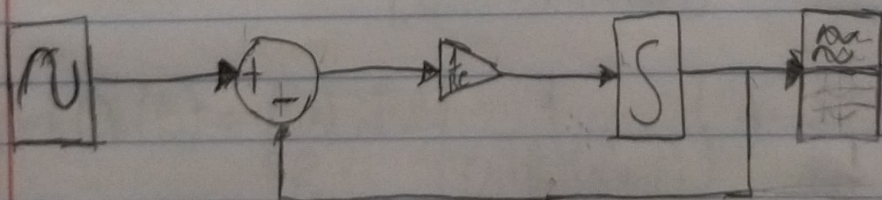
2.5.

$$g_m(\omega) = \frac{A \sin(\omega t + \phi)}{\sqrt{\omega^2 + \left(\frac{1}{RC}\right)^2}}$$

$$\Rightarrow \frac{B}{A} = \frac{1}{\sqrt{\omega^2 + \left(\frac{1}{RC}\right)^2}}$$

3.

3.1.



4.

